YUXIN CHEN

PhD · Control · Robotics · Machine Learning

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Summary

I build safe and agile embodied agents that intelligently perceive, interact with, and collaborate within the physical world while adhering to human values. My research emphasizes whole-body control of mobile robots, dexterous manipulation, and human-robot interaction, leveraging state-of-the-art advancements in reinforcement learning, generative models, optimization, and control.

Education

University of California, Berkeley — *Berkeley, CA*
PH.D. MECHANICAL ENGINEERING (CONTROL) — *Aug 2022 – May 2027*
- Advisor: Prof. Masayoshi Tomizuka
- Minors: Machine Learning, Optimization

University of Michigan, Ann Arbor — *Ann Arbor, MI*
M.S. ROBOTICS — *Aug 2020 – May 2022*
- Advisor: Prof. Ram Vasudevan

University of Michigan, Ann Arbor — *Ann Arbor, MI*
B.S.E. AEROSPACE ENGINEERING (SUMMA CUM LAUDE) — *Sep 2018 – May 2020*
- Minor: Computer Science

Shanghai Jiao Tong University — *Shanghai, China*
B.S. MECHANICAL ENGINEERING — *Sep 2016 – Aug 2020*

Research Experience

University of California, Berkeley — *Berkeley, CA*
GRADUATE STUDENT RESEARCHER — *Aug 2022 – Present*
Faculty member: Prof. Masayoshi Tomizuka
Affiliation: Mechanical Systems Control (MSC) Laboratory & Berkeley AI Research (BAIR) & Center for Humanoid Intelligence (HIC)

University of Michigan, Ann Arbor — *Ann Arbor, MI*
GRADUATE STUDENT RESEARCHER — *May 2020 – Jul 2022*
Faculty member: Prof. Ram Vasudevan
Affiliation: Robotics and Optimization for the Analysis of Human Motion (ROAHM) Laboratory

University of Michigan, Ann Arbor — *Ann Arbor, MI*
UNDERGRADUATE RESEARCH ASSISTANT — *Oct 2018 – May 2020*
Faculty member: Prof. Ella Atkins & Prof. Brent Gillespie
Affiliation: Autonomous Aerospace Systems (A2SYS) Laboratory & HAPTIX Laboratory

Working Experience

Nvidia, Corp. — *Santa Clara, CA*
RESEARCH INTERN. (MENTOR: MARCO PAVONE & BORIS IVANOVIC) — *Mar 2026 – Present*

Robotics and AI Institute — *Cambridge, MA*
RESEARCH INTERN. (MENTOR: JIUGUANG WANG) — *Mar 2025 – Aug 2025*

Mitsubishi Electric Research Laboratories — *Cambridge, MA*
RESEARCH INTERN. (MENTOR: DEVESH JHA & DIEGO ROMERES) — *May 2024 – Aug 2024*

Zoox, Inc. — *Foster City, CA*
SOFTWARE ENGINEERING INTERN. (MENTOR: RICK ZHANG) — *May 2021 – Aug 2021*

Honda R&D Americas, LLC — *Ann Arbor, MI*
STUDENT MEMBER, MULTIDISCIPLINARY DESIGN PROGRAM (MENTOR: TYLER NAES) — *Jan 2021 – Dec 2021*

ZF (China) Investment Co., Ltd — *Shanghai, China*
SOFTWARE DEVELOPMENT & TESTING INTERN. (MENTOR: YI ZHANG) — *Jan 2018 – Mar 2018*

PUBLICATIONS

*(The superscript * indicates equal contribution.)*

Journal

[J4] K. Fang*, **Y. Chen***, X. Zhu*, F. Niroui, L. Sun, J. Wang, "SAGA: Open-World Mobile Manipulation via Structured Affordance Grounding," *IEEE Robotics and Automation Letters (RA-L)*, 2026.

[J3] S. Zhao*, X. Zhu*, **Y. Chen**, C. Li, X. Zhang, M. Ding, M. Tomizuka, "DexH2R: Task-oriented Dexterous Manipulation from Human to Robots," *IEEE/ASME Transactions on Mechatronics (T-MECH)*, 2025.

[J2] R. Jalayer, **Y. Chen**, M. Jalayer, C. Orsenigo, M. Tomizuka, "Testing Human-Hand Segmentation on In-Distribution and Out-of-Distribution Data in Human-Robot Interactions Using a Deep Ensemble Model," *Mechatronics*, 2025.

[J1] P. Ewen, A. Li, **Y. Chen**, S. Hong and R. Vasudevan, "These Maps are Made for Walking: Real-Time Terrain Property Estimation for Mobile Robots," *IEEE Robotics and Automation Letters (RA-L)*, 2022.

Conference Proceedings

[C17] **Y. Chen**, H. Srikanth, N. Jew, M. Wu, P. Wang, J. Ren, M. Tomizuka, P. Xu, J. Xie, R. Tian, "CLIFT: Turning Gemini Robotics On-Device into Humanoid Specialists via Non-Invasive Closed-Loop Iterative Fine-Tuning," *2026 Conference on Robot Learning (CoRL)*, 2026. (acceptance rate: 32.8%)

[C16] A. Li*, **Y. Chen***, Z. Li, Z. Cao, J. Ren, M. Tomizuka, D. Shah, "Humanoid Navigation in Cluttered Environments with a Whole-Body Vision-Language-Action Model," *2026 Conference on Robot Learning (CoRL)*, 2026. (acceptance rate: 32.8%, **Spotlight: 3.44%**)

[C15] J. Cao*, **Y. Chen***, Y. Song, M. Tomizuka, C. Li, R. Tian, "TEXEDO: Test Time Scaling for Controller-Aware Language-Conditioned Humanoid Motion Generation," *2026 Conference on Robot Learning (CoRL)*, 2026. (acceptance rate: 32.8%)

[C14] Z. Huang, Y. Li, C. Zhang, R. Zhang, **Y. Chen**, G. Wang, X. Liu, M. Tomizuka, X. Ji, "3DWay: Generalizing Robot Manipulation via 3D Consistent Waypoints," *The 19th European Conference on Computer Vision (ECCV)*, 2026. (acceptance rate: 27.5%)

[C13] F. Zhang, P. Wang, C. Li, Y. Li, **Y. Chen**, L. Feng, C. Xu, M. Tomizuka, B. An, "REAR: Test-time Preference Realignment through Reward Decomposition," *Forty-Third International Conference on Machine Learning (ICML)*, 2025. (acceptance rate: 26.6%)

[C12] **Y. Chen***, J. Wei*, C. Xu, B. Li, M. Tomizuka, A. Bajcsy, R. Tian, "Reimagination with Test-time Observation Interventions: Distractor-Robust World Model Predictions for Visual Model Predictive Control," *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026. (acceptance rate: 38.0%)

[C11] S. Zhao, K. Yang, **Y. Chen**, C. Li, Y. Xie, X. Zhang, C. Wang, M. Tomizuka, "DexCtrl: Towards Sim-to-Real Dexterity with Adaptive Controller Learning," *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026. (acceptance rate: 38.0%)

[C10] G. Zhan, L. Tao, P. Wang, Y. Wang, Y. Li, **Y. Chen**, M. Tomizuka, S. Li, "Mean Flow Policy with Instantaneous Velocity Constraints for One-step Action Generation," *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026. (acceptance rate: 28.2%, **Oral: 1.18%**)

[C9] X. Zhu*, **Y. Chen***, L. Sun*, F. Niroui, S. Le Cleac'h, J. Wang, K. Fang, "Versatile Loco-Manipulation through Flexible Interlimb Coordination," *2025 Conference on Robot Learning (CoRL)*, 2025. (acceptance rate: 35.8%, **Oral: 5.69%**)

[C8] **Y. Chen***, C. Tang*, J. Wei, C. Li, R. Tian, X. Zhang, W. Zhan, P. Stone, M. Tomizuka, "MEReQ: Max-Ent Residual-Q Inverse RL for Sample-Efficient Alignment from intervention," *2025 Conference on Robot Learning (CoRL)*, 2025. (acceptance rate: 35.8%)

[C7] **Y. Chen**, D. Jha, M. Tomizuka, D. Romeres, "FDPP: Fine-tune Diffusion Policy with Human Preference," *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. (acceptance rate: 38.7%)

[C6] T. Zhang, Z. Wu, **Y. Chen**, Y. Wang, B. Liang, S. Moura, M. Tomizuka, M. Ding, W. Zhan, "Physics-Aware Robotic Palletization with Online Masking Inference," *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. (acceptance rate: 38.7%, **Best Paper Award in Automation: 10/4153**)

[C5] **Y. Chen***, Y. Xu*, J. Nie, Y. Wang, H. Zhuang, M. Okumura, "Advancing Cross-domain Discriminability in Continual Learning of Vision-Language Models," *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. (acceptance rate: 25.8%)

[C4] **Y. Chen**, C. Tang, R. Tian, C. Li, J. Li, M. Tomizuka and W. Zhan, "Quantifying Interaction Level Between Agents Helps Cost-efficient Generalization in Multi-agent Reinforcement Learning," *Proceedings of the 1st Reinforcement Learning Conference (RLC)*, 2024. (acceptance rate: 40%)

[C3] **Y. Chen**, C. Tang, R. Tian, C. Li, J. Li, M. Tomizuka and W. Zhan, "Quantifying Agent Interaction in Multi-Agent Reinforcement Learning for Cost-efficient Generalization," *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2024. (Extended Abstract, acceptance rate: 45.1%)

[C2] P. Ewen, J.P. Sleiman, **Y. Chen**, W.C. Lu, M. Hutter and R. Vasudevan, "Generating Continuous Motion and Force Plans in Real-Time for Legged Mobile Manipulation," *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021. (acceptance rate: 43.6%)

[C1] M. Romano, **Y. Chen**, O. Marshall, and E. Atkins, "Nailed it: Autonomous Roofing with a Nailgun-Equipped Octocopter," *AIAA Aviation Forum,* 2021.

Workshop

[W12] Y. Zeng, C. Peng, **Y. Chen**, M. Tomizuka, "RobotSTAR: Next-Scale Autoregressive Sign Language Motion Translation for Humanoid Robots," *IROS 2026 The 4th Workshop on Nonverbal Cues for Human-Robot Cooperative Intelligence*, 2026. (**Best Paper Award**)

[W11] M. Zhao*, **Y. Chen***, C. Hong, Y. Shao, J. Chen, K. Sreenath, M. Tomizuka, "Force as the Interface: Hierarchical Motion-Force Control for Force-Adaptive Humanoid Loco-manipulation," *IROS 2026 First Workshop on Sim2Real and Classical Control: From Rigorous Theory to Data-Driven Robotics*, 2026.

[W10] A. Li*, **Y. Chen***, Z. Li*, Z. Cao*, J. Ren, M. Tomizuka, D. Shah, "TANGO: Humanoid Navigation in Cluttered Environments with a Whole-Body Vision-Language-Action Model," *RSS 2026 Workshop on Whole-body Control and Bimanual Manipulation: Applications in Humanoids and Beyond (WCBM)*, 2026. (**Oral**)

[W9] M. Zhao*, **Y. Chen***, S. Zhu, Y. Hong, Y. Wang, Y. Liu, X. Tian, B. Ye, B. Jia, H. Zhao, S. Huang, M. Tomizuka, "Can We Tune Humanoid Behavior Foundation Models for Dynamic and Contact-Rich Tasks?" *RSS 2026 Workshop on Whole-body Control and Bimanual Manipulation: Applications in Humanoids and Beyond (WCBM)*, 2026.

[W8] J. Cao*, **Y. Chen***, Y. Song, M. Tomizuka, C. Li, R. Tian, "TEXEDO: Test Time Scaling for Controller-Aware Language-Conditioned Humanoid Motion Generation," *RSS 2026 Workshop on Whole-body Control and Bimanual Manipulation: Applications in Humanoids and Beyond (WCBM)*, 2026.

[W7] J. Cao*, **Y. Chen***, Y. Song, M. Tomizuka, C. Li, R. Tian, "TEXEDO: Test Time Scaling for Controller-Aware Language-Conditioned Humanoid Motion Generation," *RSS 2026 Workshop on Data-Centric Robotics: What Data Do Robots Really Need? (RoboData)*, 2026.

[W6] M. Zhao*, **Y. Chen***, S. Zhu, Y. Hong, Y. Wang, Y. Liu, X. Tian, B. Ye, B. Jia, H. Zhao, S. Huang, M. Tomizuka, "Can We Tune Humanoid Behavior Foundation Models for Dynamic and Contact-Rich Tasks?" *RSS 2026 Workshop Towards Robust Execution of Long-Horizon Whole-Body Control Task (ExWBC)*, 2026. (**Oral**)

[W5] A. Li*, **Y. Chen***, Z. Li*, Z. Cao*, J. Ren, M. Tomizuka, D. Shah, "TANGO: Humanoid Navigation in Cluttered Environments with a Whole-Body Vision-Language-Action Model," *RSS 2026 Workshop Towards Robust Execution of Long-Horizon Whole-Body Control Task (ExWBC)*, 2026.

[W4] X. Zhu*, **Y. Chen***, L. Sun*, F. Niroui, S. Le Cleac'h, J. Wang, K. Fang, "Versatile Loco-Manipulation through Flexible Interlimb Coordination," *RSS 2025 Workshop on Whole-body Control and Bimanual Manipulation: Applications in Humanoids and Beyond (WCBM)*, 2025.

[W3] **Y. Chen***, J. Wei*, C. Xu, B. Li, M. Tomizuka, A. Bajcsy, R. Tian, "Reimagination with Test-time Observation Interventions: Distractor-Robust World Model Predictions for Visual Model Predictive Control," *RSS 2025 Workshop on Out-of-Distribution Generalization in Robotics (OOD)*, 2025. (**Oral**, **Best Paper Finalist**)

[W2] **Y. Chen**, D. Jha, M. Tomizuka, D. Romeres, "Adapting Diffusion Policies to Human Preferences via Reward-Guided Fine-Tuning," *ICRA 2025 Workshop on Safely Leveraging Vision-Language Foundation Models in Robotics: Challenges and Opportunities (Safe-VLM),* 2025. (**Oral**)

[W1] P. Wang, X. Zhu, **Y. Chen**, C. Xu, M. Tomizuka, C. Li, "Residual Policy Gradient: A Reward View of KL-regularized Objective," *ICRA 2025 Workshop on Safely Leveraging Vision-Language Foundation Models in Robotics: Challenges and Opportunities (Safe-VLM),* 2025. (**Spotlight**)

Preprints and Working Papers

[P8] Y. Zeng*, C. Peng*, **Y. Chen***, A. Shao, N. Jew, M. Tomizuka, "RoboSTAR: Next-Scale Autoregressive Sign Language Translation for Humanoid Robots," *under review*, 2026.

[P7] B. Du*, Y. Xie*, S. Zhao*, **Y. Chen**, M. Wu, M. Tomizuka, "One from Infinity: Actualizing Futures from Pretrained World Models into Robot Actions," *under review*, 2026.

[P6] M. Zhao*, **Y. Chen***, Y. Shao, C. Hong, J. Chen, K. Sreenath, M. Tomizuka, "Ctrl-F: Force as an Interface for Humanoid Loco-Manipulation," *under review*, 2026.

[P5] M. Wu*, K. Yao*, S. Luan, M. Tomizuka, **Y. Chen**, "RopeFormer: Cross-Trial Adaptation from Interaction History for Dynamic Rope Manipulation," *under review*, 2026.

[P4] X. Zhang*, **Y. Chen***, Z. Liang*, G. Zhan, C. Li, C. Xu, M. Tomizuka, Y. Li, "Immiscible Diffusion Policy: Preserving Multimodal Robot Actions through Label-Free Noise Assignment," *under review*, 2026.

[P3] W. Li*, B. Yang*, **Y. Chen**, A. Wang, M. Tomizuka, "TacGoosebumps: Adding Shear Encoding to Normal-Only Tactile Sensors for Learning Contact-Rich Manipulation," *under review*, 2026.

[P2] J. Cao*, **Y. Chen***, M. Tomizuka, "CLAW: Composable Language-annotated Whole-body Motion Generation," *under review*, 2026.

[P1] Z. Xu, J. Liu, **Y. Chen**, K. Keutzer, M. Tomizuka, C. Xu, C. Peng, "Rethinking Image-to-3D Generation with Sparse Queries: Efficiency, Capacity, and Input-View Bias," *under review*, 2025.

Awards and Scholarships

2026 **Third Place in Logistic Picking (First Place in the U.S.)**, *ICRA 2026 What Bimanuals Can Do (WBCD) Competition*
2026 **Qualcomm Innovation Fellowship Finalist**, *Qualcomm Incorporated*
2025 **Departmental Block Grant Fellowship**, *University of California, Berkeley*
2025 **ICRA 2025 Best Paper Award in Automation**, *IEEE Robotics and Automation Society*
2025 **Qualcomm Innovation Fellowship Finalist**, *Qualcomm Incorporated*
2020 **Outstanding Graduates of Shanghai (top 3%)**, *Ministry of Education of Shanghai*
2020 **Capstone Design Gold Award (top 1%)**, *Shanghai Jiao Tong University*
2020 **James B. Angell Scholar**, *University of Michigan*
2019 **Roger King Scholarship**, *University of Michigan*
2018 **Longey-SJTU Global Elite Scholarship**, *Shanghai Jiao Tong University*
2017 **Rongchang Science and Technology Innovation Scholarship**, *Shanghai Jiao Tong University*
2017 **Undergraduate Academic Excellence Scholarship**, *Shanghai Jiao Tong University*

Grants

2026 **Nvidia Academic Grant Program**, *Nvidia*
2025 **Google-BAIR Commons**, *Google DeepMind*

Teaching Experience

University of California, Berkeley *Berkeley, CA*
Advanced Control System I (MECENG 232) - Graduate Student Instructor *Fall 2024*
Instructor: Prof. Masayoshi Tomizuka
Advanced Control System II (MECENG 233) - Graduate Student Instructor *Spring 2024, Spring 2026*
Instructor: Prof. Masayoshi Tomizuka
AI for Autonomy (MECENG 292B) - Graduate Student Instructor *Spring 2024*
Instructor: Dr. Wei Zhan

University of Michigan, Ann Arbor *Ann Arbor, MI*
Self-Driving Cars: Perception and Control (ROB 535) - Graduate Student Instructor *Fall 2021*
Instructor: Prof. Ram Vasudevan
Motion Planning (EECS 598) - Course Assistant *Winter 2021*
Instructor: Prof. Dmitry Berenson
Introduction to Aerospace Systems (AERO 201) - Course Assistant *Fall 2019*
Instructor: Prof. Ella Atkins

Academic Services

Journal Reviewer
- IEEE Robotics and Automation Letters (RA-L) *2024 – Present*

Conference Reviewer
- International Conference on Learning Representations (ICLR) *2025 – 2027*
- International Conference on Machine Learning (ICML) *2025 – 2026*
- Robotics: Science and Systems (RSS) *2025 – 2026*
- Conference on Robot Learning (CoRL) *2025 – 2026*
- IEEE International Conference on Robotics and Automation (ICRA) *2024 – 2026*
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) *2023 – 2026*
- Reinforcement Learning Conference (RLC) *2025 – 2026*
- Conference on Computer Vision and Pattern Recognition (CVPR) *2026*
- European Conference on Computer Vision (ECCV) *2026*
- Learning for Dynamics & Control Conference (L4DC) *2025*

Program Committee
- Co-organizer of IAVVC workshop on Scenario and Behavior Diversity in Simulation for Autonomous Vehicle Validation *2023*
- Program Committee of ICRA workshop on Human-Centered Robot Learning in the Era of Big Data and Large Models *2025*

- Program Committee for AAAI Conference on Artificial Intelligence *2025 – 2027*

Invited Talks

Advancing Cross-domain Discriminability in Continual Learning of Vision-Language Models

- TwelveLabs Webinar *Feb 2025*

Scaling Mobile Manipulation from Whole-body Coordination to Open-world Semantics

- Shenglanxueyuan Webinar *Dec 2025*
- Machine Vision and Intelligence Group at Shanghai Jiao Tong University *Jan 2026*
- Laboratory for Intelligent Decision and Autonomous Robots (LIDAR) at Georgia Institute of Technology *Feb 2026*

Bridging High-level Reasoning and Low-level Control for Loco-manipulation and Dexterity

- BAIR Humnaoid Intelligence Center Rototics Workshop *Apr 2026*

TANGO: Humanoid Navigation in Cluttered Environments with a Whole-Body Vision-Language-Action Model

- Princeton Robotic Intelligence and Systems Lab (PRISM) at Princeton University *Jun 2026*
- RoboPapers *Oct 2026*

Toward Deployable Humanoid FMs: Whole-Body Navigation, Motion Generation, and Closed-Loop Adaptation

- General Intelligent Machines *Jul 2026*

Turning Gemini Robotics On-Device into Humanoid Specialists via Non-Invasive Closed-Loop Iterative Fine-Tuning

- Lumina *Aug 2026*
- Google DeepMind *Aug 2026*

Mentorship

Yikuan Fang	Undergraduate Student (Now a Ph.D. student at Carnegie Mellon University)
Yiren Rong	Undergraduate Student (Now a Ph.D. student at Duke University)
Tianqi Zhang	Undergraduate Student (Now a Ph.D. student at The University of North Carolina at Chapel Hill)
Jianglan Wei	Undergraduate Student (Now a Master's student at Carnegie Mellon University)
Jingyuan Chen	Undergraduate Student (Now a Master's student at University of Pennsylvania)
Jianuo Cao	Undergraduate Student at Nanjing University
Hari Narayan Srikanth	Undergraduate Student at University of California, Berkeley
Nathan Alexander Jew	Undergraduate Student at University of California, Berkeley
Anqi Li	Undergraduate Student at Peking University
Zhaobo Li	Master's Student at University of California, Berkeley
Zhuo Cao	Undergraduate Student at Tsinghua University
Mengjie Zhao	Undergraduate Student at Tsinghua University
Menglin Wu	Undergraduate Student at Xi'an Jiaotong University
Jiakang Chen	Undergraduate Student at Tsinghua University
Shangbo Luan	Undergraduate Student at Peking University
Kaixiang Yao	Undergraduate Student at The Southern University of Science and Technology
Xiao Zhang	Undergraduate Student at University of California, Berkeley
Yujia Zeng	Undergraduate Student at The University of Science and Technology of China
Yuzhen Song	Undergraduate Student at The Southern University of Science and Technology
Alex Shao	Undergraduate Student at University of California, Berkeley
Akhil Agarwal	Undergraduate Student at University of California, Berkeley
Shuning Song	Undergraduate Student at Tsinghua University